
EDUCATION

University of Cincinnati

Bachelor of Science in Astrophysics and Physics, Minor in Mathematics

Cincinnati, OH

August 2021-May 2025

[Stars and Galaxies, Solar System, Intermediate Mechanics, Electromagnetism I & II, Thermostatistical Physics, Topics in Astronomy, Advanced Physics Laboratory, Introduction to Astrophysics I & II, Introduction to Quantum Mechanics I & II, Calculus I & II, Multivariable Calculus, Differential Equations, Linear Algebra, Probability and Statistics I & II]

RESEARCH EXPERIENCE

Inter-University Centre for Astronomy and Astrophysics

Post-Baccalaureate Researcher, *advised by Dr. A.N. Ramaprakash*

Pune, Maharashtra

July 2025- Present

- Commissioning and validation of WALOP's calibration accuracy using lab and simulated datasets; Stokes IQU recovery, polarization-angle zero-point, and efficiency stability
- Developed Python pipelines for dual-beam extraction, demodulation, and full uncertainty propagation on lab and simulated datasets.
- Characterizing instrumental polarization, beam-mismatch proxies, and intensity-to-polarization leakage; iterating models against measurements to reduce residuals toward survey tolerances
- Designing Fourier-based demodulation and Mueller calibration models.
- Contributing optical alignment, mechanical validation, and lab characterization of WALOP-North.

University of Cincinnati

Research Assistant, *advised by Dr. Colin Bischoff*

Cincinnati, OH

January 2024 – April 2025

- Developed Python algorithms implementing the moments expansion method to model spatial variability in CMB dust spectral indices.
- Modeled polarized dust foreground spatial and spectral properties impacting B-mode polarization measurements, using Python tools developed to analyze dust SEDs and angular power spectra.
- Analyzed the impact of spatial spectral variation on B-mode power spectra relevant to the BICEP/Keck experiment.
- Prepared framework for future integration with Bayesian parameter inference and likelihood analysis on real CMB data.

Carnegie Institution for Science (EPL)

SURI Research Intern, *advised by Dr. Dana Anderson*

Washington DC

May 2023 - August 2023

- Conducted in-depth analysis of protoplanetary disks, running simulations to investigate the impact of varied parameters on column densities and fluxes of key components.
- Simulated the chemical evolution of planet-forming materials using `rac2d` and advanced modeling techniques.
- Conducted parameterized analysis on protoplanetary disks to evaluate composition and flux variations.
- Presented data insights through visualizations, aiding in the understanding of planetary formation processes.

PRESENTATIONS

- iPoster Presentation, American Astronomical Society 244th Meeting, Madison, WI June 2024
- End-of-Summer Research Talk, Carnegie Institution for Science SURI Program, Washington, DC August 2023
- Invited Poster, University of Cincinnati Undergraduate Research Symposium, Cincinnati, OH April 2025

LEADERSHIP & ACTIVITIES

- Peer Tutor, University of Cincinnati – Learning Commons

January 2023 – April 2025

 - Provided one-on-one tutoring in Physics, Mathematics, and Chemistry.
- Peer Leader, University of Cincinnati – Learning Commons

August 2022 – December 2022

 - Led weekly workshops for first year students, facilitating collaborative problem solving, time-management coaching, and academic resource navigation.
- Arts and Science Student Ambassador, University of Cincinnati

January 2022 – December 2024

 - Represented the College of Arts and Sciences in outreach and student engagement activities to foster academic community and inclusion.
- Diversity, Equity, and Inclusion Head, Phi Mu Sorority

January 2023 – December 2023

 - Coordinated inclusive outreach events and mentorship initiatives to broaden participation.

AWARDS AND HONOURS

- Undergraduate Research Grant, University of Cincinnati
 - College of Arts & Sciences International Outreach Scholarship
 - UC Global Scholarship
 - Outstanding Tutor Award, UC Learning Center
 - Selected Participant, Carnegie SURF Program
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SKILLS

- **Programming & Data Analysis** – C++, Java, Python, MCMC, Simulation-Based Inference, Git/GitHub, Latex, MATLAB
- **Astrophysics & Data Methods** – Foreground modeling, Power spectrum estimation, Bayesian parameter inference, Cosmological and Disk simulations, FITS file handling, ds9
- **Instrumentation & Observations** – CCD imaging, Aperture photometry, Polarimetric calibration, Stokes analysis, Mueller matrix modeling, optical alignment and lab characterization for imaging polarimeters (WALOP-South/North)